

IT Infrastructure Troubleshooting Guide

A Comprehensive Knowledge Base for Common IT Issues

Version 1.0

July 28, 2025

Created for IT Administrators and End-Users

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1 Introduction

This document serves as a knowledge base for diagnosing and resolving common IT infrastructure issues. It covers a wide range of scenarios, including networking, hardware, software, security, and cloud infrastructure. Each section provides a description of the issue, potential causes, self-diagnosis steps, and recommended resolutions. This guide is intended for IT administrators, support staff, and end-users looking to troubleshoot problems efficiently.

2 Networking Issues

2.1 Unable to Connect to the Network

2.1.1 Description

Users cannot access the internet or internal network resources, such as shared drives or servers.

2.1.2 Potential Causes

- Incorrect network configuration (IP address, DNS, gateway).
- Faulty network cables or hardware (router, switch, modem).
- Wi-Fi signal interference or weak signal strength.
- Firewall or security software blocking the connection.

2.1.3 Self-Diagnosis Steps

1. Check physical connections (Ethernet cables, Wi-Fi adapter).
2. Verify IP configuration using `ipconfig` (Windows) or `ifconfig/ip addr` (Linux).
3. Ping a known server (e.g., `ping 8.8.8.8`) to test connectivity.
4. Ensure Wi-Fi is enabled and connected to the correct SSID.
5. Check firewall settings to ensure the network is not blocked.

2.1.4 Resolution

- Restart the router/modem and computer.
- Release and renew IP address (`ipconfig /release` and `ipconfig /renew` on Windows).
- Update network drivers or firmware.
- Contact the network administrator if the issue persists.

2.2 Slow Network Performance

2.2.1 Description

Network speeds are significantly slower than expected, affecting file transfers, streaming, or application performance.

2.2.2 Potential Causes

- Network congestion or high bandwidth usage.
- Outdated network drivers or firmware.
- Interference in Wi-Fi networks.
- Malware or unauthorized devices on the network.

2.2.3 Self-Diagnosis Steps

1. Run a speed test (e.g., using Speedtest.net).
2. Check for high bandwidth usage with tools like `netsat` or network monitoring software.
3. Scan for malware using antivirus software.
4. Verify Wi-Fi channel settings to avoid interference.

2.2.4 Resolution

- Limit bandwidth-heavy applications or prioritize traffic using QoS settings.
- Update router firmware and network drivers.
- Change Wi-Fi channel or frequency (2.4 GHz to 5 GHz).

- Disconnect unauthorized devices from the network.

3 Hardware Issues

3.1 Computer Fails to Boot

3.1.1 Description

The computer does not start, shows a blank screen, or displays error messages during startup.

3.1.2 Potential Causes

- Faulty power supply or battery.
- Corrupted BIOS/UEFI settings.
- Hardware failure (RAM, CPU, motherboard).
- Loose or disconnected cables.

3.1.3 Self-Diagnosis Steps

1. Check power connections and ensure the device is powered on.
2. Listen for beep codes or check diagnostic LEDs.
3. Reset BIOS/UEFI settings to default.
4. Test with a different power supply or battery (if applicable).

3.1.4 Resolution

- Reseat RAM, GPU, and other components.
- Replace faulty hardware components.
- Update or reflash BIOS/UEFI firmware.
- Contact hardware support for advanced diagnostics.

3.2 Overheating Issues

3.2.1 Description

The system shuts down unexpectedly, runs slowly, or feels excessively hot.

3.2.2 Potential Causes

- Dust buildup in fans or heatsinks.
- Failing cooling system (fans, thermal paste).
- High CPU/GPU usage.

- Poor ventilation or high ambient temperature.

3.2.3 Self-Diagnosis Steps

1. Monitor system temperatures using tools like HWMonitor or lm-sensors.
2. Inspect fans and heatsinks for dust or blockages.
3. Check CPU/GPU usage in Task Manager or top.
4. Ensure proper airflow around the device.

3.2.4 Resolution

- Clean fans and heatsinks with compressed air.
- Reapply thermal paste on CPU/GPU.
- Adjust power settings to reduce performance load.
- Place the device in a well-ventilated area.

4 Software Issues

4.1 Application Crashes or Freezes

4.1.1 Description

Applications close unexpectedly, become unresponsive, or display error messages.

4.1.2 Potential Causes

- Incompatible software versions.
- Corrupted application files.
- Insufficient system resources (RAM, CPU).
- Conflicting background processes or drivers.

4.1.3 Self-Diagnosis Steps

1. Check system resource usage in Task Manager or top.
2. Verify application compatibility with the operating system.
3. Review error logs in Event Viewer (Windows) or /var/log (Linux).
4. Test the application in Safe Mode.

4.1.4 Resolution

- Update the application to the latest version.
- Reinstall the application.

- Close unnecessary background processes.
- Update or roll back conflicting drivers.

4.2 Operating System Errors

4.2.1 Description

The OS displays errors like Blue Screen of Death (BSOD) on Windows or kernel panics on Linux/macOS.

4.2.2 Potential Causes

- Corrupted system files.
- Incompatible or faulty drivers.
- Malware infection.
- Hardware issues (e.g., failing hard drive).

4.2.3 Self-Diagnosis Steps

1. Note the error code or message displayed.
2. Run `sfc /scannow` (Windows) or `fsck` (Linux) to check system files.
3. Scan for malware using antivirus software.
4. Test hardware components (e.g., RAM with MemTest86).

4.2.4 Resolution

- Repair system files using OS recovery tools.
- Update or reinstall drivers.
- Remove malware and restore affected files.
- Replace faulty hardware components.

5 Security Issues

5.1 Malware or Virus Infection

5.1.1 Description

The system behaves erratically, displays pop-ups, or experiences unauthorized access.

5.1.2 Potential Causes

- Visiting malicious websites or downloading infected files.

- Outdated antivirus software.
- Unpatched operating system or applications.
- Phishing attacks or social engineering.

5.1.3 Self-Diagnosis Steps

1. Run a full system scan with updated antivirus software.
2. Check for unusual processes in Task Manager or top.
3. Review network activity for suspicious connections.
4. Verify recently installed software or browser extensions.

5.1.4 Resolution

- Quarantine or remove detected malware.
- Update all software and the operating system.
- Reset passwords for affected accounts.
- Enable two-factor authentication (2FA) where possible.

5.2 Unauthorized Access

5.2.1 Description

Accounts or systems are accessed without permission, or suspicious activity is detected.

5.2.2 Potential Causes

- Weak or compromised passwords.
- Lack of multi-factor authentication.
- Unsecured remote access tools (e.g., RDP, SSH).
- Insider threats or misconfigured permissions.

5.2.3 Self-Diagnosis Steps

1. Review account login history for unfamiliar locations or devices.
2. Check security logs for unauthorized access attempts.
3. Verify user permissions and active sessions.
4. Ensure remote access services are secure.

5.2.4 Resolution

- Change all affected passwords immediately.
- Enable 2FA on all accounts.
- Disable unused accounts or services.
- Implement network security measures (e.g., VPN, firewall rules).

6 Cloud Infrastructure Issues

6.1 Cloud Service Downtime

6.1.1 Description

Cloud services (e.g., AWS, Azure, GCP) are unavailable or experiencing degraded performance.

6.1.2 Potential Causes

- Provider-side outages or maintenance.
- Misconfigured cloud resources (e.g., VMs, storage).
- Network connectivity issues to the cloud provider.
- Exceeding resource quotas or limits.

6.1.3 Self-Diagnosis Steps

1. Check the cloud providers status page (e.g., AWS Service Health Dashboard).
2. Verify resource configurations in the cloud console.
3. Test connectivity to the cloud providers endpoints.
4. Review usage quotas and billing alerts.

6.1.4 Resolution

- Wait for provider-side resolution if an outage is confirmed.
- Correct misconfigured resources (e.g., scaling policies, security groups).
- Optimize resource usage to stay within quotas.
- Contact the cloud providers support if needed.

6.2 Data Sync or Access Issues

6.2.1 Description

Data stored in cloud services (e.g., Google Drive, OneDrive) is not syncing or is inaccessible.

6.2.2 Potential Causes

- Sync client issues or outdated software.
- Insufficient storage space.
- Permission or sharing configuration errors.
- Network connectivity problems.

6.2.3 Self-Diagnosis Steps

1. Verify sync client status and error messages.
2. Check available storage space in the cloud service.
3. Review file or folder sharing permissions.
4. Test network connectivity to the cloud service.

6.2.4 Resolution

- Update or reinstall the sync client.
- Free up storage space or upgrade the plan.
- Correct permission settings.
- Restart the sync process or reconnect the client.

7 Appendix

7.1 Useful Tools and Resources

headerblue	
Wireshark	Network protocol analyzer for troubleshooting network issues.
HWMonitor	Monitors hardware metrics like temperature and voltage.
MemTest86	Tests RAM for errors.
Event Viewer	Windows tool for viewing system and application logs.
SANS Cheat Sheets	Free IT and security cheat sheets. SANS Institute

7.2 Glossary

- **IP Address:** A unique address that identifies a device on a network.

- **DNS:** Domain Name System, translates domain names to IP addresses.
- **BSOD:** Blue Screen of Death, a critical error screen in Windows.
- **2FA:** Two-Factor Authentication, an additional layer of security.

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2.1.4 Resolution

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1. Run a speed test (e.g., using [Speedtest.net](https://www.speedtest.net)).
2. Check for high bandwidth usage with tools like netsat or network monitoring software.
3. Scan for malware using antivirus software.
4. Verify Wi-Fi channel settings to avoid interference.

2.2.4 Resolution

- Limit bandwidth-heavy applications or prioritize traffic using QoS settings.
- Update router firmware and network drivers.
- Change Wi-Fi channel or frequency (2.4 GHz to 5 GHz).
- Disconnect unauthorized devices from the network.

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- Hardware failure (RAM, CPU, motherboard).
- Loose or disconnected cables.

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1. Check power connections and ensure the device is powered on.
2. Listen for beep codes or check diagnostic LEDs.
3. Reset BIOS/UEFI settings to default.
4. Test with a different power supply or battery (if applicable).

3.1.4 Resolution

- Reseat RAM, GPU, and other components.
- Replace faulty hardware components.
- Update or reflash BIOS/UEFI firmware.
- Contact hardware support for advanced diagnostics.

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3.2.2 Potential Causes

- Dust buildup in fans or heatsinks.
- Failing cooling system (fans, thermal paste).
- High CPU/GPU usage.
- Poor ventilation or high ambient temperature.

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2. Inspect fans and heatsinks for dust or blockages.
3. Check CPU/GPU usage in Task Manager or top.
4. Ensure proper airflow around the device.

3.2.4 Resolution

- Clean fans and heatsinks with compressed air.

- Reapply thermal paste on CPU/GPU.
- Adjust power settings to reduce performance load.
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4.1 Application Crashes or Freezes

4.1.1 Description

Applications close unexpectedly, become unresponsive, or display error messages.

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- Incompatible software versions.
- Corrupted application files.
- Insufficient system resources (RAM, CPU).
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1. Check system resource usage in Task Manager or top.
2. Verify application compatibility with the operating system.
3. Review error logs in Event Viewer (Windows) or /var/log (Linux).
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4.1.4 Resolution

- Update the application to the latest version.
- Reinstall the application.
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The OS displays errors like Blue Screen of Death (BSOD) on Windows or kernel panics on Linux/macOS.

4.2.2 Potential Causes

- Corrupted system files.
- Incompatible or faulty drivers.
- Malware infection.

- Hardware issues (e.g., failing hard drive).

4.2.3 Self-Diagnosis Steps

1. Note the error code or message displayed.
2. Run `sfc /scannow` (Windows) or `fsck` (Linux) to check system files.
3. Scan for malware using antivirus software.
4. Test hardware components (e.g., RAM with MemTest86).

4.2.4 Resolution

- Repair system files using OS recovery tools.
- Update or reinstall drivers.
- Remove malware and restore affected files.
- Replace faulty hardware components.

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- Outdated antivirus software.
- Unpatched operating system or applications.
- Phishing attacks or social engineering.

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2. Check for unusual processes in Task Manager or `top`.
3. Review network activity for suspicious connections.
4. Verify recently installed software or browser extensions.

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- Update all software and the operating system.
- Reset passwords for affected accounts.
- Enable two-factor authentication (2FA) where possible.

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5.2.1 Description

Accounts or systems are accessed without permission, or suspicious activity is detected.

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- Weak or compromised passwords.
- Lack of multi-factor authentication.
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- Insider threats or misconfigured permissions.

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2. Check security logs for unauthorized access attempts.
3. Verify user permissions and active sessions.
4. Ensure remote access services are secure.

5.2.4 Resolution

- Change all affected passwords immediately.
- Enable 2FA on all accounts.
- Disable unused accounts or services.
- Implement network security measures (e.g., VPN, firewall rules).

6 Cloud Infrastructure Issues

6.1 Cloud Service Downtime

6.1.1 Description

Cloud services (e.g., AWS, Azure, GCP) are unavailable or experiencing degraded performance.

6.1.2 Potential Causes

- Provider-side outages or maintenance.
- Misconfigured cloud resources (e.g., VMs, storage).
- Network connectivity issues to the cloud provider.
- Exceeding resource quotas or limits.

6.1.3 Self-Diagnosis Steps

1. Check the cloud providers status page (e.g., AWS Service Health Dashboard).

2. Verify resource configurations in the cloud console.
3. Test connectivity to the cloud providers endpoints.
4. Review usage quotas and billing alerts.

6.1.4 Resolution

- Wait for provider-side resolution if an outage is confirmed.
- Correct misconfigured resources (e.g., scaling policies, security groups).
- Optimize resource usage to stay within quotas.
- Contact the cloud providers support if needed.

6.2 Data Sync or Access Issues

6.2.1 Description

Data stored in cloud services (e.g., Google Drive, OneDrive) is not syncing or is inaccessible.

6.2.2 Potential Causes

- Sync client issues or outdated software.
- Insufficient storage space.
- Permission or sharing configuration errors.
- Network connectivity problems.

6.2.3 Self-Diagnosis Steps

1. Verify sync client status and error messages.
2. Check available storage space in the cloud service.
3. Review file or folder sharing permissions.
4. Test network connectivity to the cloud service.

6.2.4 Resolution

- Update or reinstall the sync client.
- Free up storage space or upgrade the plan.
- Correct permission settings.
- Restart the sync process or reconnect the client.

7 Appendix

7.1 Useful Tools and Resources

headerblue	Description
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Tool/Resource	
Wireshark	Network protocol analyzer for troubleshooting network issues.
HWMonitor	Monitors hardware metrics like temperature and voltage.
MemTest86	Tests RAM for errors.
Event Viewer	Windows tool for viewing system and application logs.
SANS Cheat Sheets	Free IT and security cheat sheets. SANS Institute

7.2 Glossary

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- Firewall or security software blocking the connection.

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6. Check physical connections (Ethernet cables, Wi-Fi adapter).
7. Verify IP configuration using `ipconfig` (Windows) or `ifconfig/ip addr` (Linux).
8. Ping a known server (e.g., `ping 8.8.8.8`) to test connectivity.
9. Ensure Wi-Fi is enabled and connected to the correct SSID.
10. Check firewall settings to ensure the network is not blocked.

2.1.4 Resolution

- Restart the router/modem and computer.
- Release and renew IP address (`ipconfig /release` and `ipconfig /renew` on Windows).
- Update network drivers or firmware.
- Contact the network administrator if the issue persists.

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Network speeds are significantly slower than expected, affecting file transfers, streaming, or application performance.

2.2.2 Potential Causes

- Network congestion or high bandwidth usage.
- Outdated network drivers or firmware.
- Interference in Wi-Fi networks.
- Malware or unauthorized devices on the network.

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7. Scan for malware using antivirus software.

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- Limit bandwidth-heavy applications or prioritize traffic using QoS settings.
- Update router firmware and network drivers.
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- Faulty power supply or battery.
- Corrupted BIOS/UEFI settings.
- Hardware failure (RAM, CPU, motherboard).
- Loose or disconnected cables.

3.1.3 Self-Diagnosis Steps

5. Check power connections and ensure the device is powered on.
6. Listen for beep codes or check diagnostic LEDs.
7. Reset BIOS/UEFI settings to default.
8. Test with a different power supply or battery (if applicable).

3.1.4 Resolution

- Reseat RAM, GPU, and other components.
- Replace faulty hardware components.
- Update or reflash BIOS/UEFI firmware.
- Contact hardware support for advanced diagnostics.

3.2 Overheating Issues

3.2.1 Description

The system shuts down unexpectedly, runs slowly, or feels excessively hot.

3.2.2 Potential Causes

- Dust buildup in fans or heatsinks.
- Failing cooling system (fans, thermal paste).
- High CPU/GPU usage.
- Poor ventilation or high ambient temperature.

3.2.3 Self-Diagnosis Steps

3. Monitor system temperatures using tools like HWMonitor or lm-sensors.
4. Inspect fans and heatsinks for dust or blockages. 3. Check CPU/GPU usage in Task Manager or top.
4. Ensure proper airflow around the device.

3.2.4 Resolution

- Clean fans and heatsinks with compressed air.
- Reapply thermal paste on CPU/GPU.
- Adjust power settings to reduce performance load.
- Place the device in a well-ventilated area.

4 Software Issues

4.1 Application Crashes or Freezes

4.1.1 Description

Applications close unexpectedly, become unresponsive, or display error messages.

4.1.2 Potential Causes

- Incompatible software versions.
- Corrupted application files.
- Insufficient system resources (RAM, CPU).
- Conflicting background processes or drivers.

4.1.3 Self-Diagnosis Steps

5. Check system resource usage in Task Manager or top.
6. Verify application compatibility with the operating system.
7. Review error logs in Event Viewer (Windows) or /var/log (Linux).
8. Test the application in Safe Mode.

4.1.4 Resolution

- Update the application to the latest version.

- Reinstall the application.
- Close unnecessary background processes.
- Update or roll back conflicting drivers.

4.2 Operating System Errors

4.2.1 Description

The OS displays errors like Blue Screen of Death (BSOD) on Windows or kernel panics on Linux/macOS.

4.2.2 Potential Causes

- Corrupted system files.
- Incompatible or faulty drivers.
- Malware infection.
- Hardware issues (e.g., failing hard drive).

4.2.3 Self-Diagnosis Steps

5. Note the error code or message displayed.
6. Run `sfc /scannow` (Windows) or `fsck` (Linux) to check system files.
7. Scan for malware using antivirus software.
8. Test hardware components (e.g., RAM with MemTest86).

4.2.4 Resolution

- Repair system files using OS recovery tools.
- Update or reinstall drivers.
- Remove malware and restore affected files.
- Replace faulty hardware components.

5 Security Issues

5.1 Malware or Virus Infection

5.1.1 Description

The system behaves erratically, displays pop-ups, or experiences unauthorized access.

5.1.2 Potential Causes

- Visiting malicious websites or downloading infected files.
- Outdated antivirus software.
- Unpatched operating system or applications.

- Phishing attacks or social engineering.

5.1.3 Self-Diagnosis Steps

5. Run a full system scan with updated antivirus software.
6. Check for unusual processes in Task Manager or top.
7. Review network activity for suspicious connections.
8. Verify recently installed software or browser extensions.

5.1.4 Resolution

- Quarantine or remove detected malware.
- Update all software and the operating system.
- Reset passwords for affected accounts.
- Enable two-factor authentication (2FA) where possible.

5.2 Unauthorized Access

5.2.1 Description

Accounts or systems are accessed without permission, or suspicious activity is detected.

5.2.2 Potential Causes

- Weak or compromised passwords.
- Lack of multi-factor authentication.
- Unsecured remote access tools (e.g., RDP, SSH).
- Insider threats or misconfigured permissions.

5.2.3 Self-Diagnosis Steps

5. Review account login history for unfamiliar locations or devices.
6. Check security logs for unauthorized access attempts.
7. Verify user permissions and active sessions.
8. Ensure remote access services are secure.

5.2.4 Resolution

- Change all affected passwords immediately.
- Enable 2FA on all accounts.
- Disable unused accounts or services.
- Implement network security measures (e.g., VPN, firewall rules).

6 Cloud Infrastructure Issues

6.1 Cloud Service Downtime

6.1.1 Description

Cloud services (e.g., AWS, Azure, GCP) are unavailable or experiencing degraded performance.

6.1.2 Potential Causes

- Provider-side outages or maintenance.
- Misconfigured cloud resources (e.g., VMs, storage).
- Network connectivity issues to the cloud provider.
- Exceeding resource quotas or limits.

6.1.3 Self-Diagnosis Steps

5. Check the cloud providers status page (e.g., AWS Service Health Dashboard).
6. Verify resource configurations in the cloud console.
7. Test connectivity to the cloud providers endpoints.
8. Review usage quotas and billing alerts.

6.1.4 Resolution

- Wait for provider-side resolution if an outage is confirmed.
- Correct misconfigured resources (e.g., scaling policies, security groups).
- Optimize resource usage to stay within quotas.
- Contact the cloud providers support if needed.

6.2 Data Sync or Access Issues

6.2.1 Description

Data stored in cloud services (e.g., Google Drive, OneDrive) is not syncing or is inaccessible.

6.2.2 Potential Causes

- Sync client issues or outdated software.
- Insufficient storage space.
- Permission or sharing configuration errors.
- Network connectivity problems.

6.2.3 Self-Diagnosis Steps

5. Verify sync client status and error messages.

6. Check available storage space in the cloud service.
7. Review file or folder sharing permissions.
8. Test network connectivity to the cloud service.

6.2.4 Resolution

- Update or reinstall the sync client.
- Free up storage space or upgrade the plan.
- Correct permission settings.
- Restart the sync process or reconnect the client.

7 Appendix

7.1 Useful Tools and Resources

headerblue Tool/Resource	Description
Wireshark	Network protocol analyzer for troubleshooting network issues.
HWMonitor	Monitors hardware metrics like temperature and voltage.
MemTest86	Tests RAM for errors.
Event Viewer	Windows tool for viewing system and application logs.
SANS Cheat Sheets	Free IT and security cheat sheets. SANS Institute

7.2 Glossary

- **IP Address:** A unique address that identifies a device on a network.
- **DNS:** Domain Name System, translates domain names to IP addresses.
- **BSOD:** Blue Screen of Death, a critical error screen in Windows.
- **2FA:** Two-Factor Authentication, an additional layer of security.

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1 Introduction

This document serves as a knowledge base for diagnosing and resolving common IT infrastructure issues. It covers a wide range of scenarios, including networking, hardware, software, security, and cloud infrastructure. Each section provides a description of the issue, potential causes, self-diagnosis steps, and recommended resolutions. This guide is intended for IT administrators, support staff, and end-users looking to troubleshoot problems efficiently.

2 Networking Issues

2.1 Unable to Connect to the Network

2.1.1 Description

Users cannot access the internet or internal network resources, such as shared drives or servers.

2.1.2 Potential Causes

- Incorrect network configuration (IP address, DNS, gateway).
- Faulty network cables or hardware (router, switch, modem).
- Wi-Fi signal interference or weak signal strength.
- Firewall or security software blocking the connection.

2.1.3 Self-Diagnosis Steps

11. Check physical connections (Ethernet cables, Wi-Fi adapter).
12. Verify IP configuration using `ipconfig` (Windows) or `ifconfig/ip addr` (Linux).
13. Ping a known server (e.g., `ping 8.8.8.8`) to test connectivity.
14. Ensure Wi-Fi is enabled and connected to the correct SSID.
15. Check firewall settings to ensure the network is not blocked.

2.1.4 Resolution

- Restart the router/modem and computer.
- Release and renew IP address (`ipconfig /release` and `ipconfig /renew` on Windows).
- Update network drivers or firmware.
- Contact the network administrator if the issue persists.

2.2 Slow Network Performance

2.2.1 Description

Network speeds are significantly slower than expected, affecting file transfers, streaming, or application performance.

2.2.2 Potential Causes

- Network congestion or high bandwidth usage.
- Outdated network drivers or firmware.
- Interference in Wi-Fi networks.
- Malware or unauthorized devices on the network.

2.2.3 Self-Diagnosis Steps

9. Run a speed test (e.g., using [Speedtest.net](https://www.speedtest.net)).
10. Check for high bandwidth usage with tools like netsat or network monitoring software.
11. Scan for malware using antivirus software.
12. Verify Wi-Fi channel settings to avoid interference.

2.2.4 Resolution

- Limit bandwidth-heavy applications or prioritize traffic using QoS settings.
- Update router firmware and network drivers.
- Change Wi-Fi channel or frequency (2.4 GHz to 5 GHz).
- Disconnect unauthorized devices from the network.

3 Hardware Issues

3.1 Computer Fails to Boot

3.1.1 Description

The computer does not start, shows a blank screen, or displays error messages during startup.

3.1.2 Potential Causes

- Faulty power supply or battery.
- Corrupted BIOS/UEFI settings.
- Hardware failure (RAM, CPU, motherboard).
- Loose or disconnected cables.

3.1.3 Self-Diagnosis Steps

9. Check power connections and ensure the device is powered on.
10. Listen for beep codes or check diagnostic LEDs.

11. Reset BIOS/UEFI settings to default.
12. Test with a different power supply or battery (if applicable).

3.1.4 Resolution

- Reseat RAM, GPU, and other components.
- Replace faulty hardware components.
- Update or reflash BIOS/UEFI firmware.
- Contact hardware support for advanced diagnostics.

3.2 Overheating Issues

3.2.1 Description

The system shuts down unexpectedly, runs slowly, or feels excessively hot.

3.2.2 Potential Causes

- Dust buildup in fans or heatsinks.
- Failing cooling system (fans, thermal paste).
- High CPU/GPU usage.
- Poor ventilation or high ambient temperature.

3.2.3 Self-Diagnosis Steps

5. Monitor system temperatures using tools like HWMonitor or lm-sensors.
 6. Inspect fans and heatsinks for dust or blockages. 3. Check CPU/GPU usage in Task Manager or top.
4. Ensure proper airflow around the device.

3.2.4 Resolution

- Clean fans and heatsinks with compressed air.
- Reapply thermal paste on CPU/GPU.
- Adjust power settings to reduce performance load.
- Place the device in a well-ventilated area.

4 Software Issues

4.1 Application Crashes or Freezes

4.1.1 Description

Applications close unexpectedly, become unresponsive, or display error messages.

4.1.2 Potential Causes

- Incompatible software versions.
- Corrupted application files.
- Insufficient system resources (RAM, CPU).
- Conflicting background processes or drivers.

4.1.3 Self-Diagnosis Steps

9. Check system resource usage in Task Manager or top.
10. Verify application compatibility with the operating system.
11. Review error logs in Event Viewer (Windows) or /var/log (Linux).
12. Test the application in Safe Mode.

4.1.4 Resolution

- Update the application to the latest version.
- Reinstall the application.
- Close unnecessary background processes.
- Update or roll back conflicting drivers.

4.2 Operating System Errors

4.2.1 Description

The OS displays errors like Blue Screen of Death (BSOD) on Windows or kernel panics on Linux/macOS.

4.2.2 Potential Causes

- Corrupted system files.
- Incompatible or faulty drivers.
- Malware infection.
- Hardware issues (e.g., failing hard drive).

4.2.3 Self-Diagnosis Steps

9. Note the error code or message displayed.
10. Run sfc /scannow (Windows) or fsck (Linux) to check system files.
11. Scan for malware using antivirus software.
12. Test hardware components (e.g., RAM with MemTest86).

4.2.4 Resolution

- Repair system files using OS recovery tools.

- Update or reinstall drivers.
- Remove malware and restore affected files.
- Replace faulty hardware components.

5 Security Issues

5.1 Malware or Virus Infection

5.1.1 Description

The system behaves erratically, displays pop-ups, or experiences unauthorized access.

5.1.2 Potential Causes

- Visiting malicious websites or downloading infected files.
- Outdated antivirus software.
- Unpatched operating system or applications.
- Phishing attacks or social engineering.

5.1.3 Self-Diagnosis Steps

9. Run a full system scan with updated antivirus software.
10. Check for unusual processes in Task Manager or top.
11. Review network activity for suspicious connections.
12. Verify recently installed software or browser extensions.

5.1.4 Resolution

- Quarantine or remove detected malware.
- Update all software and the operating system.
- Reset passwords for affected accounts.
- Enable two-factor authentication (2FA) where possible.

5.2 Unauthorized Access

5.2.1 Description

Accounts or systems are accessed without permission, or suspicious activity is detected.

5.2.2 Potential Causes

- Weak or compromised passwords.
- Lack of multi-factor authentication.
- Unsecured remote access tools (e.g., RDP, SSH).
- Insider threats or misconfigured permissions.

5.2.3 Self-Diagnosis Steps

9. Review account login history for unfamiliar locations or devices.
10. Check security logs for unauthorized access attempts.
11. Verify user permissions and active sessions.
12. Ensure remote access services are secure.

5.2.4 Resolution

- Change all affected passwords immediately.
- Enable 2FA on all accounts.
- Disable unused accounts or services.
- Implement network security measures (e.g., VPN, firewall rules).

6 Cloud Infrastructure Issues

6.1 Cloud Service Downtime

6.1.1 Description

Cloud services (e.g., AWS, Azure, GCP) are unavailable or experiencing degraded performance.

6.1.2 Potential Causes

- Provider-side outages or maintenance.
- Misconfigured cloud resources (e.g., VMs, storage).
- Network connectivity issues to the cloud provider.
- Exceeding resource quotas or limits.

6.1.3 Self-Diagnosis Steps

9. Check the cloud providers status page (e.g., AWS Service Health Dashboard).
10. Verify resource configurations in the cloud console.
11. Test connectivity to the cloud providers endpoints.
12. Review usage quotas and billing alerts.

6.1.4 Resolution

- Wait for provider-side resolution if an outage is confirmed.
- Correct misconfigured resources (e.g., scaling policies, security groups).
- Optimize resource usage to stay within quotas.
- Contact the cloud providers support if needed.

6.2 Data Sync or Access Issues

6.2.1 Description

Data stored in cloud services (e.g., Google Drive, OneDrive) is not syncing or is inaccessible.

6.2.2 Potential Causes

- Sync client issues or outdated software.
- Insufficient storage space.
- Permission or sharing configuration errors.
- Network connectivity problems.

6.2.3 Self-Diagnosis Steps

9. Verify sync client status and error messages.
10. Check available storage space in the cloud service.
11. Review file or folder sharing permissions.
12. Test network connectivity to the cloud service.

6.2.4 Resolution

- Update or reinstall the sync client.
- Free up storage space or upgrade the plan.
- Correct permission settings.
- Restart the sync process or reconnect the client.

7 Appendix

7.1 Useful Tools and Resources

headerblue Tool/Resource	Description
Wireshark	Network protocol analyzer for troubleshooting network issues.
HWMonitor	Monitors hardware metrics like temperature and voltage.
MemTest86	Tests RAM for errors.
Event Viewer	Windows tool for viewing system and application logs.
SANS Cheat Sheets	Free IT and security cheat sheets. SANS Institute

7.2 Glossary

- **IP Address:** A unique address that identifies a device on a network.
- **DNS:** Domain Name System, translates domain names to IP addresses.
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- **2FA:** Two-Factor Authentication, an additional layer of security.

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1 Introduction

This document serves as a knowledge base for diagnosing and resolving common IT infrastructure issues. It covers a wide range of scenarios, including networking, hardware, software, security, and cloud infrastructure. Each section provides a description of the issue, potential causes, self-diagnosis steps, and recommended resolutions. This guide is intended for IT administrators, support staff, and end-users looking to troubleshoot problems efficiently.

2 Networking Issues

2.1 Unable to Connect to the Network

2.1.1 Description

Users cannot access the internet or internal network resources, such as shared drives or servers.

2.1.2 Potential Causes

- Incorrect network configuration (IP address, DNS, gateway).
- Faulty network cables or hardware (router, switch, modem).
- Wi-Fi signal interference or weak signal strength.
- Firewall or security software blocking the connection.

2.1.3 Self-Diagnosis Steps

16. Check physical connections (Ethernet cables, Wi-Fi adapter).
17. Verify IP configuration using `ipconfig` (Windows) or `ifconfig/ip addr` (Linux).

18. Ping a known server (e.g., ping 8.8.8.8) to test connectivity.
19. Ensure Wi-Fi is enabled and connected to the correct SSID.
20. Check firewall settings to ensure the network is not blocked.

2.1.4 Resolution

- Restart the router/modem and computer.
- Release and renew IP address (ipconfig /release and ipconfig /renew on Windows).
- Update network drivers or firmware.
- Contact the network administrator if the issue persists.

2.2 Slow Network Performance

2.2.1 Description

Network speeds are significantly slower than expected, affecting file transfers, streaming, or application performance.

2.2.2 Potential Causes

- Network congestion or high bandwidth usage.
- Outdated network drivers or firmware.
- Interference in Wi-Fi networks.
- Malware or unauthorized devices on the network.

2.2.3 Self-Diagnosis Steps

13. Run a speed test (e.g., using [Speedtest.net](https://www.speedtest.net)).
14. Check for high bandwidth usage with tools like netsat or network monitoring software.
15. Scan for malware using antivirus software.
16. Verify Wi-Fi channel settings to avoid interference.

2.2.4 Resolution

- Limit bandwidth-heavy applications or prioritize traffic using QoS settings.
- Update router firmware and network drivers.
- Change Wi-Fi channel or frequency (2.4 GHz to 5 GHz).
- Disconnect unauthorized devices from the network.

3 Hardware Issues

3.1 Computer Fails to Boot

3.1.1 Description

The computer does not start, shows a blank screen, or displays error messages during startup.

3.1.2 Potential Causes

- Faulty power supply or battery.
- Corrupted BIOS/UEFI settings.
- Hardware failure (RAM, CPU, motherboard).
- Loose or disconnected cables.

3.1.3 Self-Diagnosis Steps

13. Check power connections and ensure the device is powered on.
14. Listen for beep codes or check diagnostic LEDs.
15. Reset BIOS/UEFI settings to default.
16. Test with a different power supply or battery (if applicable).

3.1.4 Resolution

- Reseat RAM, GPU, and other components.
- Replace faulty hardware components.
- Update or reflash BIOS/UEFI firmware.
- Contact hardware support for advanced diagnostics.

3.2 Overheating Issues

3.2.1 Description

The system shuts down unexpectedly, runs slowly, or feels excessively hot.

3.2.2 Potential Causes

- Dust buildup in fans or heatsinks.
- Failing cooling system (fans, thermal paste).
- High CPU/GPU usage.
- Poor ventilation or high ambient temperature.

3.2.3 Self-Diagnosis Steps

7. Monitor system temperatures using tools like HWMonitor or lm-sensors.
8. Inspect fans and heatsinks for dust or blockages. 3. Check CPU/GPU usage in Task Manager or top.
4. Ensure proper airflow around the device.

3.2.4 Resolution

- Clean fans and heatsinks with compressed air.
- Reapply thermal paste on CPU/GPU.
- Adjust power settings to reduce performance load.
- Place the device in a well-ventilated area.

4 Software Issues

4.1 Application Crashes or Freezes

4.1.1 Description

Applications close unexpectedly, become unresponsive, or display error messages.

4.1.2 Potential Causes

- Incompatible software versions.
- Corrupted application files.
- Insufficient system resources (RAM, CPU).
- Conflicting background processes or drivers.

4.1.3 Self-Diagnosis Steps

13. Check system resource usage in Task Manager or top.
14. Verify application compatibility with the operating system.
15. Review error logs in Event Viewer (Windows) or /var/log (Linux).
16. Test the application in Safe Mode.

4.1.4 Resolution

- Update the application to the latest version.
- Reinstall the application.
- Close unnecessary background processes.
- Update or roll back conflicting drivers.

4.2 Operating System Errors

4.2.1 Description

The OS displays errors like Blue Screen of Death (BSOD) on Windows or kernel panics on Linux/macOS.

4.2.2 Potential Causes

- Corrupted system files.
- Incompatible or faulty drivers.

- Malware infection.
- Hardware issues (e.g., failing hard drive).

4.2.3 Self-Diagnosis Steps

13. Note the error code or message displayed.
14. Run sfc /scannow (Windows) or fsck (Linux) to check system files.
15. Scan for malware using antivirus software.
16. Test hardware components (e.g., RAM with MemTest86).

4.2.4 Resolution

- Repair system files using OS recovery tools.
- Update or reinstall drivers.
- Remove malware and restore affected files.
- Replace faulty hardware components.

5 Security Issues

5.1 Malware or Virus Infection

5.1.1 Description

The system behaves erratically, displays pop-ups, or experiences unauthorized access.

5.1.2 Potential Causes

- Visiting malicious websites or downloading infected files.
- Outdated antivirus software.
- Unpatched operating system or applications.
- Phishing attacks or social engineering.

5.1.3 Self-Diagnosis Steps

13. Run a full system scan with updated antivirus software.
14. Check for unusual processes in Task Manager or top.
15. Review network activity for suspicious connections.
16. Verify recently installed software or browser extensions.

5.1.4 Resolution

- Quarantine or remove detected malware.
- Update all software and the operating system.
- Reset passwords for affected accounts.

- Enable two-factor authentication (2FA) where possible.

5.2 Unauthorized Access

5.2.1 Description

Accounts or systems are accessed without permission, or suspicious activity is detected.

5.2.2 Potential Causes

- Weak or compromised passwords.
- Lack of multi-factor authentication.
- Unsecured remote access tools (e.g., RDP, SSH).
- Insider threats or misconfigured permissions.

5.2.3 Self-Diagnosis Steps

13. Review account login history for unfamiliar locations or devices.
14. Check security logs for unauthorized access attempts.
15. Verify user permissions and active sessions.
16. Ensure remote access services are secure.

5.2.4 Resolution

- Change all affected passwords immediately.
- Enable 2FA on all accounts.
- Disable unused accounts or services.
- Implement network security measures (e.g., VPN, firewall rules).

6 Cloud Infrastructure Issues

6.1 Cloud Service Downtime

6.1.1 Description

Cloud services (e.g., AWS, Azure, GCP) are unavailable or experiencing degraded performance.

6.1.2 Potential Causes

- Provider-side outages or maintenance.
- Misconfigured cloud resources (e.g., VMs, storage).
- Network connectivity issues to the cloud provider.
- Exceeding resource quotas or limits.

6.1.3 Self-Diagnosis Steps

13. Check the cloud providers status page (e.g., AWS Service Health Dashboard).
14. Verify resource configurations in the cloud console.
15. Test connectivity to the cloud providers endpoints.
16. Review usage quotas and billing alerts.

6.1.4 Resolution

- Wait for provider-side resolution if an outage is confirmed.
- Correct misconfigured resources (e.g., scaling policies, security groups).
- Optimize resource usage to stay within quotas.
- Contact the cloud providers support if needed.

6.2 Data Sync or Access Issues

6.2.1 Description

Data stored in cloud services (e.g., Google Drive, OneDrive) is not syncing or is inaccessible.

6.2.2 Potential Causes

- Sync client issues or outdated software.
- Insufficient storage space.
- Permission or sharing configuration errors.
- Network connectivity problems.

6.2.3 Self-Diagnosis Steps

13. Verify sync client status and error messages.
14. Check available storage space in the cloud service.
15. Review file or folder sharing permissions.
16. Test network connectivity to the cloud service.

6.2.4 Resolution

- Update or reinstall the sync client.
- Free up storage space or upgrade the plan.
- Correct permission settings.
- Restart the sync process or reconnect the client.

7 Appendix

7.1 Useful Tools and Resources

headerblue Tool/Resource	Description
Wireshark	Network protocol analyzer for troubleshooting network issues.
HWMonitor	Monitors hardware metrics like temperature and voltage.
MemTest86	Tests RAM for errors.
Event Viewer	Windows tool for viewing system and application logs.
SANS Cheat Sheets	Free IT and security cheat sheets. SANS Institute

7.2 Glossary

- **IP Address:** A unique address that identifies a device on a network.
 - **DNS:** Domain Name System, translates domain names to IP addresses.
 - **BSOD:** Blue Screen of Death, a critical error screen in Windows.
 - **2FA:** Two-Factor Authentication, an additional layer of security.
17. Check the cloud providers status page (e.g., AWS Service Health Dashboard).
 18. Verify resource configurations in the cloud console.
 19. Test connectivity to the cloud providers endpoints.
 20. Review usage quotas and billing alerts.

6.1.4 Resolution

- Wait for provider-side resolution if an outage is confirmed.
- Correct misconfigured resources (e.g., scaling policies, security groups).
- Optimize resource usage to stay within quotas.
- Contact the cloud providers support if needed.

6.2 Data Sync or Access Issues

6.2.1 Description

Data stored in cloud services (e.g., Google Drive, OneDrive) is not syncing or is inaccessible.

6.2.2 Potential Causes

- Sync client issues or outdated software.
- Insufficient storage space.

- Permission or sharing configuration errors.
- Network connectivity problems.

6.2.3 Self-Diagnosis Steps

17. Verify sync client status and error messages.
18. Check available storage space in the cloud service.
19. Review file or folder sharing permissions.
20. Test network connectivity to the cloud service.

6.2.4 Resolution

- Update or reinstall the sync client.
- Free up storage space or upgrade the plan.
- Correct permission settings.
- Restart the sync process or reconnect the client.

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21. Check the cloud providers status page (e.g., AWS Service Health Dashboard).
22. Verify resource configurations in the cloud console.
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